Pocket Money Academy

Pocket Money Academy *(student proposal)*

> **Assessment:** ◆ Fit with adjustments · **Category:** Education + Training · **Assessed:** 2026-07-10 · **Status:** awaiting instructor approval

Proposal (as submitted): A fintech app that appeals to 6–12 year olds, teaching the fundamentals of financial management.

Category note: proposed as Fintech, but the product's core promise is **teaching** — learners (kids) and authors (parents/teachers) are the first-class roles, so it belongs in **Education + Training** (runner-up: Fintech). Fintech's signature demands (precision, high-frequency feeds, auditability) are not what this app is about; an interactive teaching engine is.

Adjustments required for approval:

- The teaching core must be a **load-bearing C++ simulated pocket-money economy** — allowance accrual, chore rewards, shop prices, savings interest and life events ticking in a rendered, animated 2D world — not quiz screens or form-based lessons. Without this, the real-time loop is decorative and the project collapses into CRUD.
- **Virtual money only** — no real banking, cards or payment rails (that is licensed-fintech territory, cf. Greenlight/Acorns Early); the module has no path to real-money handling.
- **Child privacy by design** — parent-created accounts, no child emails/real names/PII in Firebase, COPPA-style data minimization (Bankaroo's model).
- The parent/teacher role must be first-class: the M2 content editor authors missions, shop catalogs and **age-banded scenarios (6–8 / 9–12)** — the engineering must outweigh the lesson library.

References:

- Bankaroo — virtual bank for kids (official)
- Greenlight — debit card & money app for kids (official)

- Acorns Early, formerly GoHenry (official)

Core Tech Requirements

Legend: ✔ Applies directly | ◆ Applies with domain adaptation | ● Optional | ✘ Not needed | » Second trimester

Core Tech Requirement	Pocket Money Academy
External Database (Firebase)	✔
Authentication »	✔
Analytics »	✔
C++ Performant Service	✔
Real-time C++ Simulation Loop	◆
Application Communication	✔
Continuous Integration	✔
Data-Driven Architecture	✔
Front-end + Local Backend	✔
Custom Tools Development	✔
2D Graphical Output	✔
3D (strong students only)	✘
Networking System	●
Localization & Content Pipeline — multi-language content with hot-reload (new)	✔

How each requirement applies:

> *Platform fit:* **Native mobile (reference device: Google Pixel 8a / 9a) + Native PC (authoring)** — kids play touch-first on a phone; the parent/teacher mission & economy editor runs on the PC authoring side of the role split.

Front end: Animated 2D town/shop view (sprite characters, coin/jar animations), savings-goal visualizations, kid-readable custom-rendered charts, icon-first reading-light UI. **Languages:** C++, GLSL.

Back end: Pocket-money economy engine (allowance scheduler, price drift, compound interest, chore/event system), mission & assessment system, Firebase progress and family sync. **Languages:** C++.
Backend scope: Local now, server-ready (T2) — a classroom edition (teacher-hosted shared class economy, cf. Bankaroo for Schools) could lift the economy service onto the T2 server.

- *Simulation loop:* fixed-timestep economy — allowance accrues, shop prices drift, savings earn compound interest, chores and life events (birthday money, broken toy) fire — driving the animated town the kid lives in.
- *Data-driven:* missions, shop items, price/interest tables, events and age-band curricula (6–8 / 9–12) all defined as JSON data.
- *Custom tool:* mission & economy editor — parents/teachers author missions, shop catalogs and scenario parameters.
- *External DB:* progress, savings goals and badges synced per kid. *Auth »:* parent accounts owning PII-free kid profiles. *Analytics »:* learning insights per money concept (saving, budgeting, interest).
- *2D:* animated town/shop rendering plus custom kid-readable charts (money jars, goal thermometers) — no chart libraries. *3D:* not needed. *Networking:* optional — family fund transfers ride on Firebase.
- *Localization pipeline:* kid content is text- and audio-heavy — language packs and age-band packs are hot-swappable data, never code.

Suggested Tech Goals

G1–G3 ★ are the common goals (fixed for all categories); the other 3 are picked from the Education + Training pool in `milestone-goals-pool-per-category.md` (G1–G3 ★ common, G4+ picks).

M1:

- **G1 ★ Application Shell, Real-time C++ Simulation Loop, Input & Core Logic**
  Tech & dependencies
  - **Tech:** SDL2 window/touch/mouse input with a fixed-timestep loop driving the economy tick and town animation; time acceleration (day/week fast-forward) for demos; C++17/20, CMake, Git.
  - **Prerequisites:** None — foundation of this project.
  - **Feeds into:** Every other goal; M1 G11's economy and M1 G9's UI states tick inside this loop.
- **G2 ★ Data-Driven Architecture of Core Objects / Entity System**
  Tech & dependencies
  - **Tech:** Kid profile, accounts, shop items, missions, events and age-band curricula fully described in JSON (nlohmann/json) with hot-reload for rapid content tuning.
  - **Prerequisites:** G1 (loop & update hooks).
  - **Feeds into:** M1 G8 (mission format), M1 G11 (economy parameters), M2 G2 (editor edits this data), purple localization pipeline.
- **G3 ★ Debugging, Profiling, Stress Test & CI**
  Tech & dependencies
  - **Tech:** ImGui overlays (economy totals, event queue, tick ms), Tracy profiling, Catch2 stress tests fast-forwarding years of simulated allowance/interest/events in seconds with balance-invariant checks, GitHub Actions Windows+Linux matrix.
  - **Prerequisites:** G1 (loop to instrument).
  - **Feeds into:** M2 G3 (CI hardening); the determinism checks that make savings math trustworthy for kids and parents.
- **G11 Domain Simulation Core** — the pocket-money economy itself: allowance scheduler, chore rewards, price drift, compound interest, life events — the load-bearing sim required by adjustment 1.

Tech & dependencies

  - **Tech:** Per-tick economy state machine (earn/save/spend/donate flows, interest compounding, event generator) with seeded RNG so a scenario replays identically.
  - **Prerequisites:** G1 (fixed timestep), G2 (economy parameters as data).
  - **Feeds into:** All money lessons; M1 G8 missions evaluate against this economy; M2 G12 rewards hook its events.

- **G8 Lesson / Exercise Content System** — missions ("save for the bike", "budget the week", "wants vs. needs at the shop") as a data-driven curriculum with 6–8 / 9–12 age bands.
  Tech & dependencies
  - **Tech:** Mission schema (goal, constraints, success criteria, age band, reward) with validation and a progression graph unlocking concepts in order.
  - **Prerequisites:** G2 (data model), M1 G11 (economy to evaluate missions against).
  - **Feeds into:** M2 G2 (editor authors missions), M2 G7 (progress per mission), M2 G12 (badges on completion).
- **G9 Learner UI / UX System** — reading-light, icon-first flows for 6–8 and richer flows for 9–12; celebration and feedback states kids understand.
  Tech & dependencies
  - **Tech:** Screen-flow system (town, shop, savings jar, missions) with age-band layouts, large touch targets, audio cues and FreeType text sized for early readers.
  - **Prerequisites:** G1 (input & render loop), G2 (age bands as data).
  - **Feeds into:** M2 G4 (visual polish), M2 G14-style mobile ergonomics validated in M2 G1.

M2:

- **G1 ★ Architecture — Optimization & Platform Validation**
  Tech & dependencies
  - **Tech:** Tracy-guided optimization of economy tick and sprite batching; ASan on MSVC; showcase build validated on the reference phone (Pixel 8a/9a) for touch latency, battery and 60 fps town animation.
  - **Prerequisites:** All M1 goals (hardens the M1 codebase).

    - **Feeds into:** The mobile showcase build every other M2 goal demos on.
- **G2 ★ Content Editor — Core** — the mission & economy editor: parents/teachers author missions, shop catalogs, price/interest tables and age-band scenarios.
  Tech & dependencies
    - **Tech:** Desktop authoring tool (PC, authoring side of the role split) editing missions, shop items and economy tables with JSON round-trip, undo/redo command stack and live preview in the project's town view.
    - **Libraries/Software:** Dear ImGui, nlohmann/json, project's SDL2/OpenGL town renderer (live preview)
    - **Prerequisites:** M1 G2 (data model), M1 G8 (mission schema), M1 G11 (economy parameters).
    - **Feeds into:** Every authored scenario; makes the parent/teacher role first-class (adjustment 4).
- **G3 ★ CI — CMake, Code Quality, Build Standards & Packaging**
  Tech & dependencies
    - **Tech:** CMake presets, clang-format + clang-tidy gates, packaging of the mobile showcase build and the PC authoring tool, GitHub Actions matrix with an economy-determinism smoke test.
    - **Prerequisites:** M1 G3 (CI and test foundation).
    - **Feeds into:** Release/showcase builds; guards M1 G11's seeded reproducibility.
- **G4 2D Graphics — Advanced** — animated money jars, goal thermometers, celebration effects and kid-readable custom charts: the visual language that makes money concepts land at age 6.
  Tech & dependencies
    - **Tech:** Sprite animation system (coin flights, jar fills, confetti), custom-rendered progress/history charts with kid-friendly scales, GPU batching for the town scene.
    - **Prerequisites:** M1 G9 (UI flows), M2 G1 (mobile frame budget).
    - **Feeds into:** The showcase demo's wow factor; M2 G12's reward moments.
- **G7 Firebase — Accounts, Progress Sync & Teacher Dashboards** — the parent dashboard: progress per money concept, goal history and mission results, on PII-free kid profiles.

Tech & dependencies

- **Tech:** Firebase Auth for parents owning nickname-only kid profiles (COPPA-style minimization), Firestore sync of progress/goals/badges, parent dashboard views.
- **Prerequisites:** M1 G8 (mission results to sync), M1 G11 (economy state snapshots).
- **Feeds into:** Adjustment 3 compliance; the T2 classroom-edition seam (backend scope).

- **G12 Gamification Systems** — badges, streaks and goal celebrations: the engagement engine that keeps 6–12 year olds practicing money habits.
  Tech & dependencies
  - **Tech:** Badge/streak rules engine driven by economy and mission events, reward animations, collection screen; all badge definitions as data.
  - **Prerequisites:** M1 G11 (economy events), M1 G8 (mission completions), M2 G4 (celebration visuals).
  - **Feeds into:** Retention story at the showcase; M2 G7 syncs earned badges to the parent dashboard.

Track Goals & Team Composition

Recommended composition: 6 CS + 1 Design + 1 Art — a children's learning product has a mid-to-high art surface (mascot, exercise scenes); a 2nd artist is justified if the direction confirms animation-heavy mascot work (max 9, D+A ≤ 3).

> **Track goals** — suggested Design/Art picks and TEAM notes for this project; deliverables & quality ladders live in the Design, Art and TEAM pool documents.
>
> - **Design M1:**
>   - D1 ★ Features DD — money-habit objectives for 6–12-year-olds, kid + parent/teacher personas, child-privacy mandate
>   - D2 Prototype — one exercise loop (earn → save/spend → reflect) playable end to end
> - **Design M2:**
>   - D1 ★ Features Design Iteration & Showcase Readiness — FDD updated from kid/parent playtests, changes documented, showcase lesson flow designed
>   - D6 Complete lesson unit — one pocket-money lesson polished end to end (intro → exercises → assessment → results)
> - **UI/UX M1:**
>   - U1 ★ UI Standards & Wireframe Baseline — child-first flows plus separate parent/teacher entry
> - **UI/UX M2:**
>   - U1 ★ Functional UI & Usability — Education screens: home/lesson select · in-exercise with pause · Resume/Quit (exact labels, confirmation) · results/progress · settings · how-to-play · separate child and parent/teacher entry points
>   - U2 ★ UX Validation & Final Polished UI — validated with real learners where possible; final polish of the child-first UI
> - **Art M1:**
>   - A1 ★ Age-appropriate visual language (mockups: 2 exercise screens)
>   - A2 Mascot — a money-buddy character with expression sheet (neutral / encouraging / corrective)

- **Art M2:**
  - A1 ★ Exercise scenes, mascot set and feedback visuals in the showcase lesson
  - A2 Mascot reaction set — integrated with the feedback/celebration pass
- **TEAM M1:**
  - TA1 ★ Overall Audio Experience — initial BGM/SFX from the week-5 needs list present and working in the prototype, suitable and correctly triggered for this project.
  - TA2 ★ Audio Engine Functionality — data-driven audio pipeline (add or replace audio without recompiling) and reliable multi-channel playback.
- **TEAM M2:**
  - TA1 ★ Overall Audio Experience — Education floor: ≥1 BGM (focus-friendly, non-distracting) + ≥6 SFX with distinct correct/incorrect/progress cues mandatory
  - TA2 ★ Audio Engine Functionality — audio properties de/serialised and the full playback functionality the build needs (multi-channel; advanced where the project calls for it).
  - TP1 ★ Final Presentation — 7 min in front of class + instructors: live custom-tools showcase, prototype demo, team post-mortem (wk14)

Generated by the Project Proposal Assessor skill · grounding: live folder · references verified 2026-07-10 · track goals & unified workbooks retrofitted 2026-07-12.